CANALRUNG · PRODUCT REQUIREMENTS AND SPECIFICATIONS

CanalRung product requirements and specifications

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DOCUMENT CONTROL

Document ID	CNR-PRS-001	Version	0.5
Status	Draft	Date	2026-11-05
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/03-requirements.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-11-05	Amish Chadha	Kit 1.9.1: identifiers R1 to R10 become PR-001 to PR-010 (column Was); each traced to a user requirement in CNR-URS-001; PR-011 to PR-013 added; the document ID CNR-REQ-001 becomes CNR-PRS-001 (older records still say CNR-REQ-001); no target changed
0.4	2026-10-03	Amish Chadha	Amish's requirement decisions of 2026-10-03 (4A and 9A, CNR-DDR-003): R2 mass target restated to 4.5 kg; R6 restated for the 500 x 1,200 mm footing as the standard at every site, in drained or saturated soil; R4 and R9 results updated for the 9 mm ropes and larger footing
0.3	2026-10-03	Amish Chadha	TRL 3: status of every requirement from CNR-CAL-001 on the constructable design (CNR-DDR-002); R2 not met on mass; R9 reported against the value-engineering target
0.2	2026-10-03	Amish Chadha	

			TRL 2: measurable targets; R1 restated for a stated design envelope; R4 and R6 given margins; R6 restated for the fixed station anchor; R10 added (nothing thrown can hurt the person in the water); decided under Amish's 2026-10-03 pre-approval (CNR-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

CanalRung product requirements and specifications

CanalRung meets nine of its ten requirements on paper or by design at TRL 3 and is over its value-engineering target on the tenth (R9). On 2026-10-03 Amish decided "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A": for CanalRung, 4A (smaller ropes, and R2 restated to 4.5 kg if the ladder still exceeds 4.0 kg) and 9A (the 500 x 1,200 mm footing standard at every site). Four requirements also need trials with people before they can be closed (R2 throw accuracy, R5 climb time, R7 deploy time, R8 ageing); those are TRL 4 work. Status comes from the calculation note CNR-CAL-001; the tag in brackets is the line of `docs/04-calcs/sizing.py` that carries the figure.

> **Safety:** CanalRung is safety-critical rescue and rope equipment. These requirements describe a research prototype, published as an open engineering reference and never as certified life-saving equipment. Meeting them on paper does not make the ladder safe to use; only proof loads, tank trials under lifeguard supervision and field trials can show that. Never enter the water to rescue someone: throw, anchor and call the emergency number.

Design envelope

The first version is sized for lined canals where the water is up to 2.0 m below the coping on a 1.5:1 (horizontal to vertical) side slope, or up to 3.0 m below the coping on a vertical wall, with surface currents up to 2.0 m/s and water from 5 to 30 °C. The design case in the model is the 1.5:1 slope with the water 2.0 m below the coping. Deeper slopes need a longer ladder (a later variant, CNR-DDR-001, D1).

Table 1. Product requirements and specifications. Status at TRL 3 from CNR-CAL-001 v0.3 (`python docs/04-calcs/sizing.py` *).*

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Reach the water from the bank	Rung section at least 4.0 m; at least two rungs at or below the water line on the design walls (1.5:1 slope to 2.0 m, vertical wall to 3.0 m)	4.0	min	m	Calculation [A1] to [A5]; deploy on a test wall at TRL 4	Met on paper: 4,020 mm; two rungs in the water on the design slope and four on a 3.0 m vertical wall; works on 1.5:1 slopes to 2.1 m and vertical walls to 3.8 m	R1
PR-002	UR-007	Usability	Throwable by one adult	Ladder as thrown (everything that leaves the box) at most 4.5 kg (9.9 lb), restated by Amish on 2026-10-03 from 4.0 kg; 8 of 10 untrained adults land the throw weight within 1 m of a target 3 m upstream	4.5	max	kg	Weigh; throw trials with volunteers (TRL 4)	Met on paper on mass: 4.27 kg [B3], 0.23 kg inside; throw accuracy needs trials	R2
PR-003	UR-008	Performance	Rungs float at the water line	Every rung floats with at least 0.15 kg to spare, with its share of rope; the bottom three rungs carry the throw weight with a factor of at least 1.5	0.15 and 1.5	min	kg; ratio	Calculation [C1] to [C3]; tank test (TRL 4)	Met on paper: 0.27 kg (plain) and 0.22 kg (stand-off) per rung; factor 1.75	R3

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-004	UR-004	Safety	Carry a climbing adult	Each rung takes 1.5 kN at mid-span at no more than two thirds of yield; the two side ropes with their knots, wet, hold at least four times 4.5 kN (18 kN)	1.5 and 18	min	kN	Calculation [D1], [E1]; proof-load test (TRL 4)	Met on paper: rung at 0.56 of yield; 9 mm ropes 18.4 kN, 4.08 times 4.5 kN	R4
PR-005	UR-003	Usability	Climbable on a flat wall	Clear width at least 280 mm; rung pitch 300 to 350 mm; at least 60 mm of hand room behind the rung at the stand-offs; an adult volunteer climbs out of a 1.5 m deep test tank with a smooth vertical wall in under 60 s	280; 60	min	mm	Geometry [A6]; supervised tank trials with lifeguards (TRL 4)	Geometry met by design (298 mm, 335 mm, 70 mm); climb time needs trials	R5
PR-006	UR-005	Safety	Anchor holds	Station anchor (eye nut, bolt, post and the standard 500 x 1,200 mm footing used at every site) holds 4.5 kN with a factor of at least 1.5 on each element, in drained or saturated granular bank soil (restated by Amish on 2026-10-03)	4.5	min	kN	Calculation [G1] to [G6]; pull test on site (TRL 4)	Met on paper on level ground and in line: eye nut 1.53; footing 3.58 drained, 1.59 saturated; the slope-crest reduction of the footing and the eye nut pull off its axis are	R6

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
									not yet checked (CNR-STM-001 S4, S5, S23)	
PR-007	UR-002	Performance	Deploy quickly	Lid open to ladder in the water in under 30 s, using only the pictogram on the lid	30	max	s	Timed trials with untrained volunteers (TRL 4)	Not verifiable at TRL 3; estimate 15 s [H1]	R7
PR-008	UR-009	Materials	Survive outdoor storage	Ropes and rungs keep at least 80 % of strength after the equivalent of 2 years of outdoor exposure	80	min	%	Accelerated UV ageing and pull test (TRL 4)	Met by design (stored in an opaque UV-stabilised box) [H2]; to confirm by test	R8
PR-009	UR-010	Cost	Value for money	Parts cost reported against the value-engineering target of USD 500, including the post, footing and box	500	max	USD	Costed bill of materials [I1]	Over target: estimated USD 605.20, USD 105.20 over	R9
PR-010	UR-006	Safety	Nothing thrown can hurt the person in the water	Soft throw weight; no exposed sharp edges or tube ends; a blunt-tip knife at the station to cut rope if anyone is caught	Pass or fail	Pass or fail	n/a	Inspection of the model and BOM [H3]	Met by design	R10
PR-011	UR-011	Physical architecture	Built and mended locally	Every made part is cut, drilled or cast from stock sections with hand tools, a drill and a	No welding, lathe or mill	Pass or fail	n/a	Inspection of the BOM, the model and the build plan BLD-001	Met by design	New

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				bandsaw or hacksaw; no welding, lathe or mill is needed for option A (the current design); bought parts are catalogue items						
PR-012	UR-002, UR-012	Usability	Obvious and inspectable station	The lid carries a pictogram how-to; a numbered tamper tag is fitted to the latch; the box is bright yellow or orange; the front decal says do not enter the water and call the emergency number	Pass or fail	Pass or fail	n/a	Inspection of the model and BOM lines 8 and 11	Met by design	New
PR-013	UR-013	Regulatory	Honest labelling	Every drawing, render and the README say "research and educational prototype, not certified life-saving equipment" and "CONCEPT, NOT FOR FABRICATION"; no standard compliance or approval is claimed	Pass or fail	Pass or fail	n/a	Inspection of the repository; `python .kit/render.py --check`	Met by design	New

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software, cost, regulatory. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4. "Was" is the identifier of the same requirement in earlier records (CNR-DDR-001 to CNR-DDR-003, CNR-CAL-001, `docs/REVIEW.md`).

Requirements not met or at risk

Older notes below name requirements R1 to R10; they are PR-001 to PR-010 (column Was).

- **R2 (restated, now met on mass):** with 9 mm ropes the ladder as thrown weighs 4.27 kg, still 0.27 kg over the earlier 4.0 kg, so the target was restated to 4.5 kg under Amish's fallback in decision 4A (CNR-DDR-003). The stronger rungs were kept. Whether 4.3 kg is easy to throw is a question for the TRL 4 throw trials.
- **R4 (thin margin on the ropes):** the 9 mm EN 1891 type B ropes hold 18.4 kN wet with knots, 4.08 times 4.5 kN, just over the four times R4 asks for and above Amish's floor of three times. The bought rope must declare at least 18 kN.
- **R6 (soil outside the screening values):** the standard footing holds with factors of 3.58 in drained and 1.59 in saturated granular soil. Soft clay or peat banks need a site pull test.
- **R9 (over the target):** the larger standard footing adds USD 65.00 of concrete; the estimate is USD 105.20 over the value-engineering target.

Assumptions

- Many canal falls happen near places where a station can be placed: crossings, pump houses, paths and farms.
- A person in cold water can grip a rung long enough to climb if the rung is within reach.
- Throwing upstream and letting the current carry the ladder is easier for a bystander than throwing to the person.
- Irrigation districts would allow stations on their banks.
- The design climber is 100 kg with a dynamic factor of 1.5 (1.47 kN); the 4.5 kN anchor target is about three times that.